

Qiskit 101 — Lab 02

Multi-Qubit Computing & Entanglement
Theory Talk → Hands-on Lab

From one qubit to many

01

State space

Why n qubits
require 2^n
amplitudes

02

Operations

Basis preparation
and CNOT

03

Entanglement

Bell states,
correlation,
non-separability

04

Scale

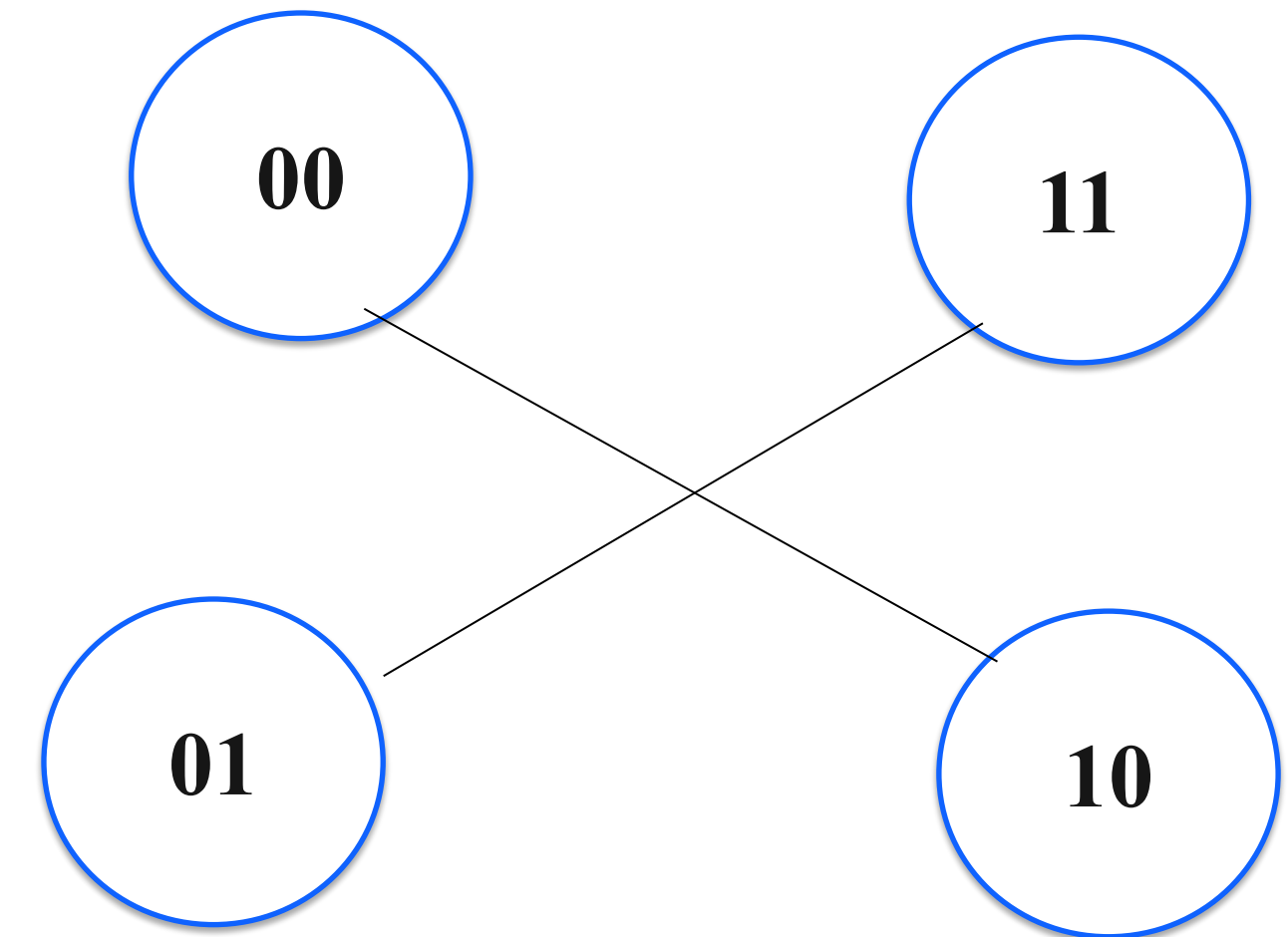
GHZ states and
circuit design

Two-qubit state space

$|00\rangle$ $|01\rangle$ $|10\rangle$ $|11\rangle$

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$$



**2 qubits → 4
basis states**

Two ideas to
keep in mind

QISKIT ORDERING

Displayed bit string uses q_1q_0

So $X(q_0)$ on $|00\rangle$ gives $|01\rangle$.

MEASUREMENT

- Amplitudes determine probabilities.
- A shot returns one classical bit string.

STATE-SPACE GROWTH

- $1 \rightarrow 2$ amplitudes
- $2 \rightarrow 4$
- $3 \rightarrow 8$
- $n \rightarrow 2^n$

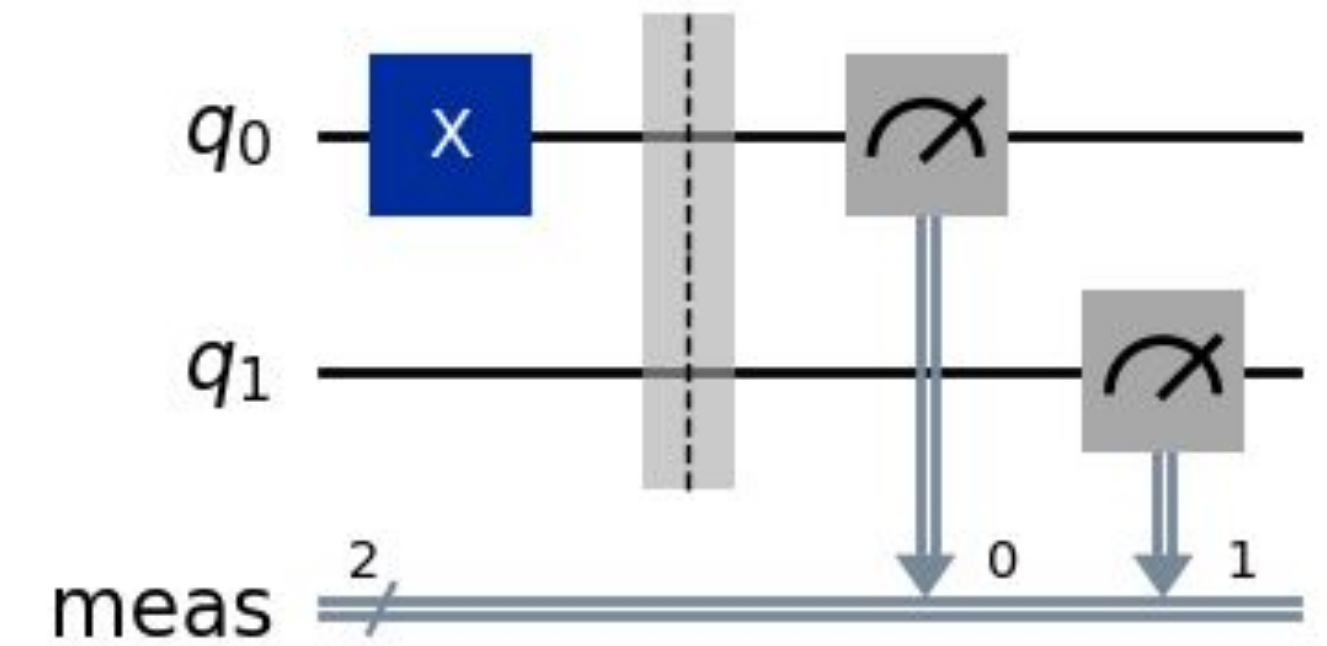
Prepare basis states
before adding control

Start: $|00\rangle$

$X(q_0) \rightarrow |01\rangle$

$X(q_1) \rightarrow |10\rangle$

$X(q_0), X(q_1) \rightarrow |11\rangle$



Qiskit

```
qc = QuantumCircuit(2)
```

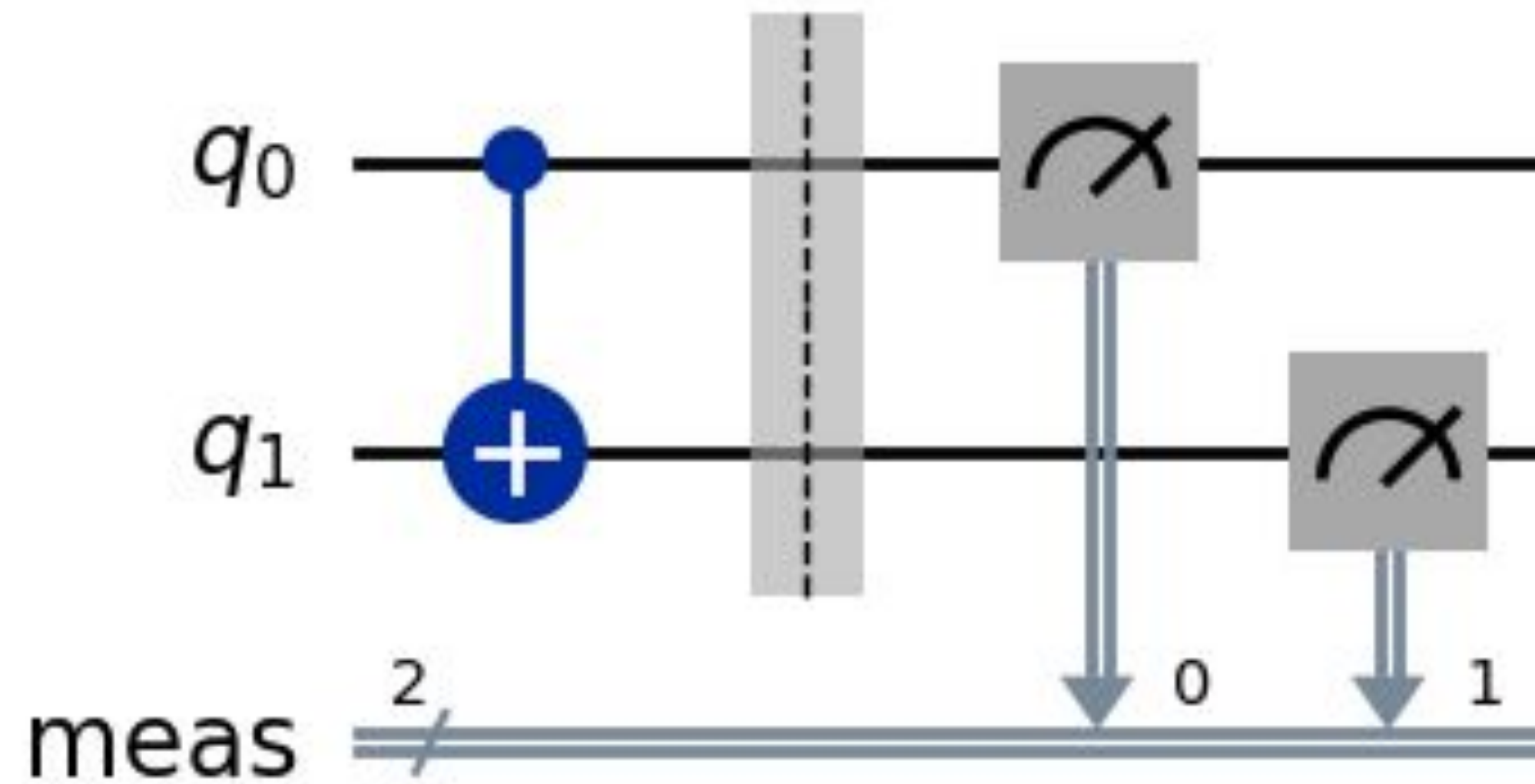
```
qc.x(0)
```

```
qc.measure_all()
```

**prepare $\rightarrow$ transform $\rightarrow$
measure**

Controlled operations connect qubits

CNOT: the target flips only when the control is $|1\rangle$.



CNOT: one qubit controls another (q1, q0)

00 → 00

01 → 11

10 → 10

11 → 01

CONTROL q_0

- $q_0 = 0 \rightarrow$ target unchanged
- $q_0 = 1 \rightarrow$ target X is applied

TARGET q_1

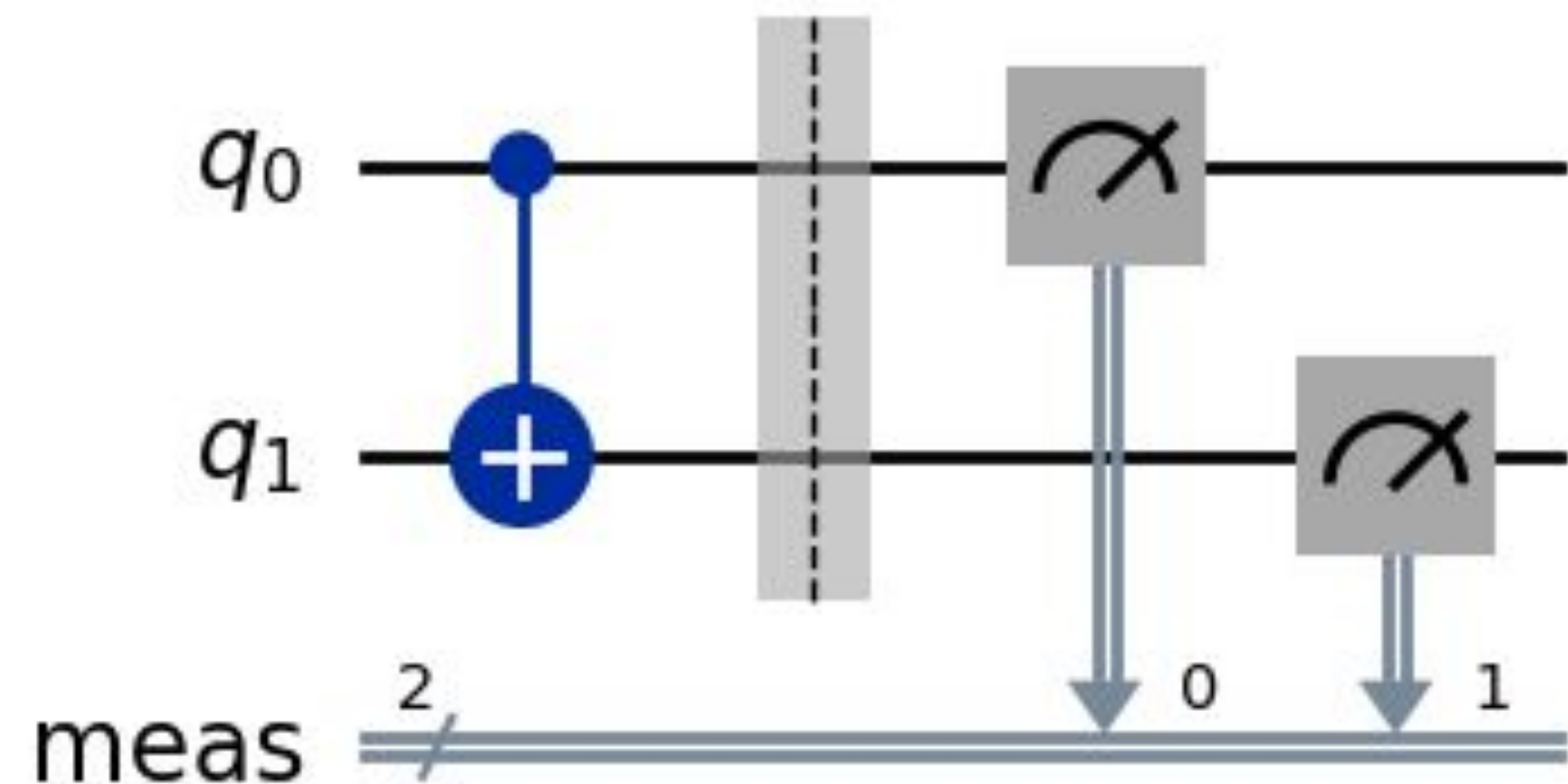
qc.cx(0, 1)

$|00\rangle \rightarrow |00\rangle$

$|01\rangle \rightarrow |11\rangle$

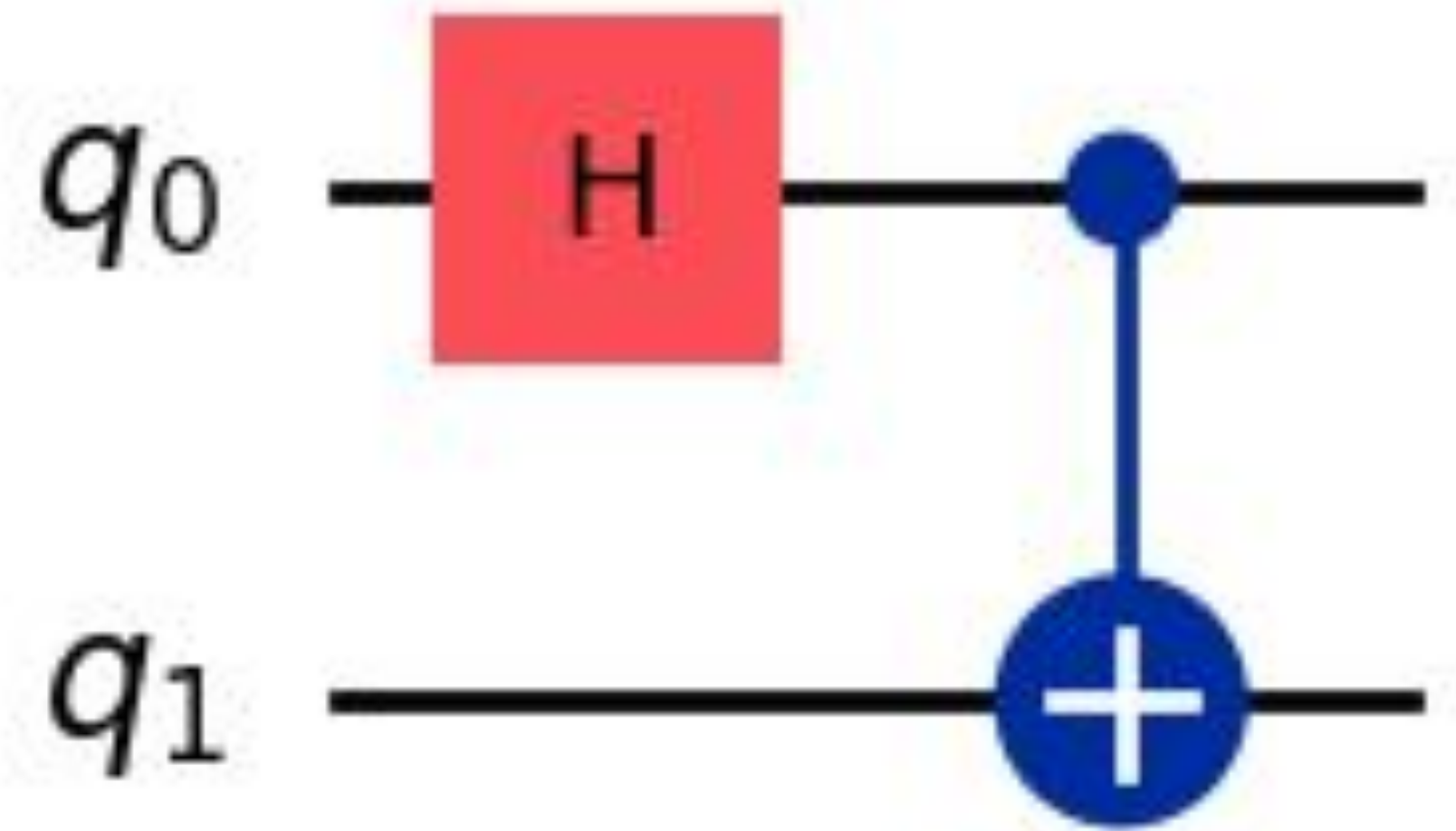
$|10\rangle \rightarrow |10\rangle$

$|11\rangle \rightarrow |01\rangle$



H + CNOT creates the first entangled state

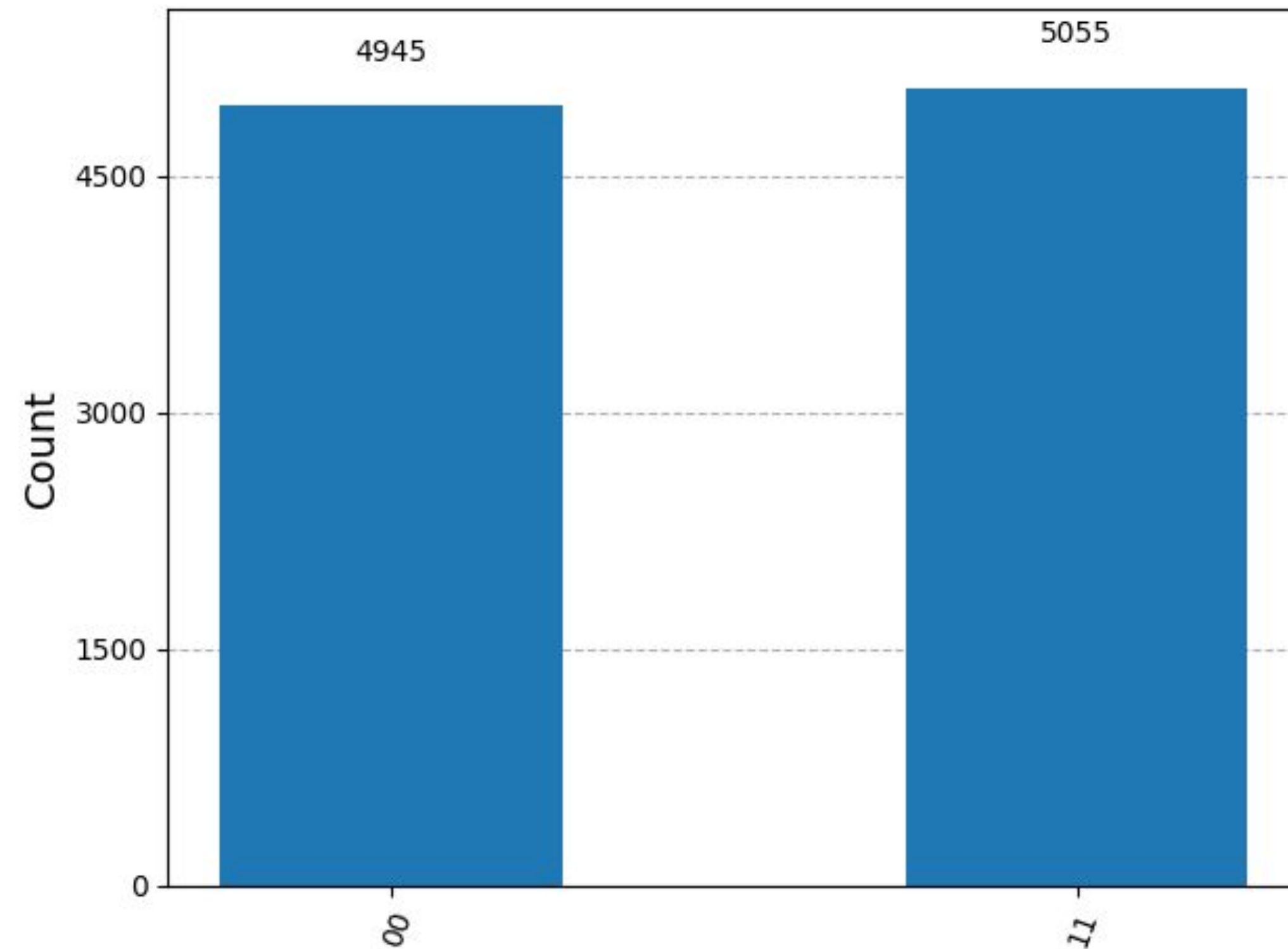
$$\begin{aligned} &|00\rangle \\ &\downarrow H(q_0) \\ &(|00\rangle + |01\rangle)/\sqrt{2} \\ &\downarrow CX(q_0, q_1) \\ &|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2} \end{aligned}$$



Predict first: only 00 and 11 should be observed.

Bell-state measurement: random outcomes, fixed relationship

- $P(00)=1/2$ $P(11)=1/2$
- $P(01)=0$ $P(10)=0$
- The individual shot is unpredictable.
- The relationship is deterministic: $q_0 = q_1$.



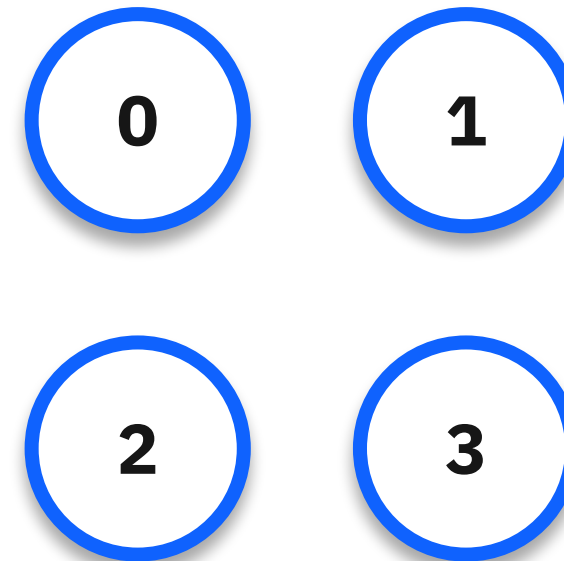
**Superposition is not
the same as
entanglement**

PRODUCT STATE

$$|+\rangle|+\rangle$$

$$= (|00\rangle + |01\rangle + |10\rangle + |11\rangle)/2$$

Four outcomes $\approx 25\%$ each



product

ENTANGLED STATE

$$|\Phi^+\rangle$$

$$= (|00\rangle + |11\rangle)/\sqrt{2}$$

Only 00 and 11 occur

Non-separable



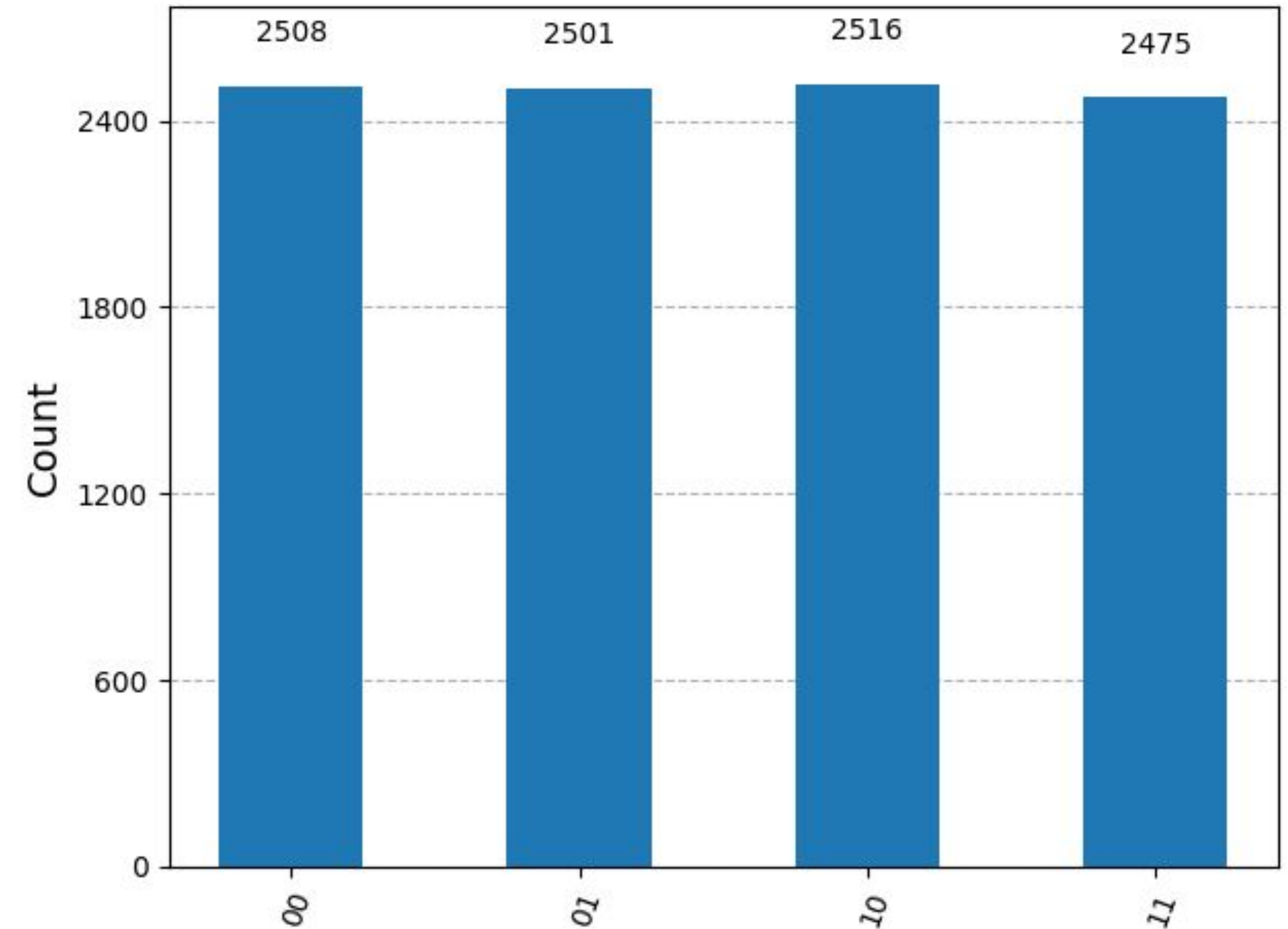
entangled

Statevectors reveal structure

Product: $[1/2 \ 1/2 \ 1/2 \ 1/2]^T$

Bell: $[1/\sqrt{2} \ 0 \ 0 \ 1/\sqrt{2}]^T$

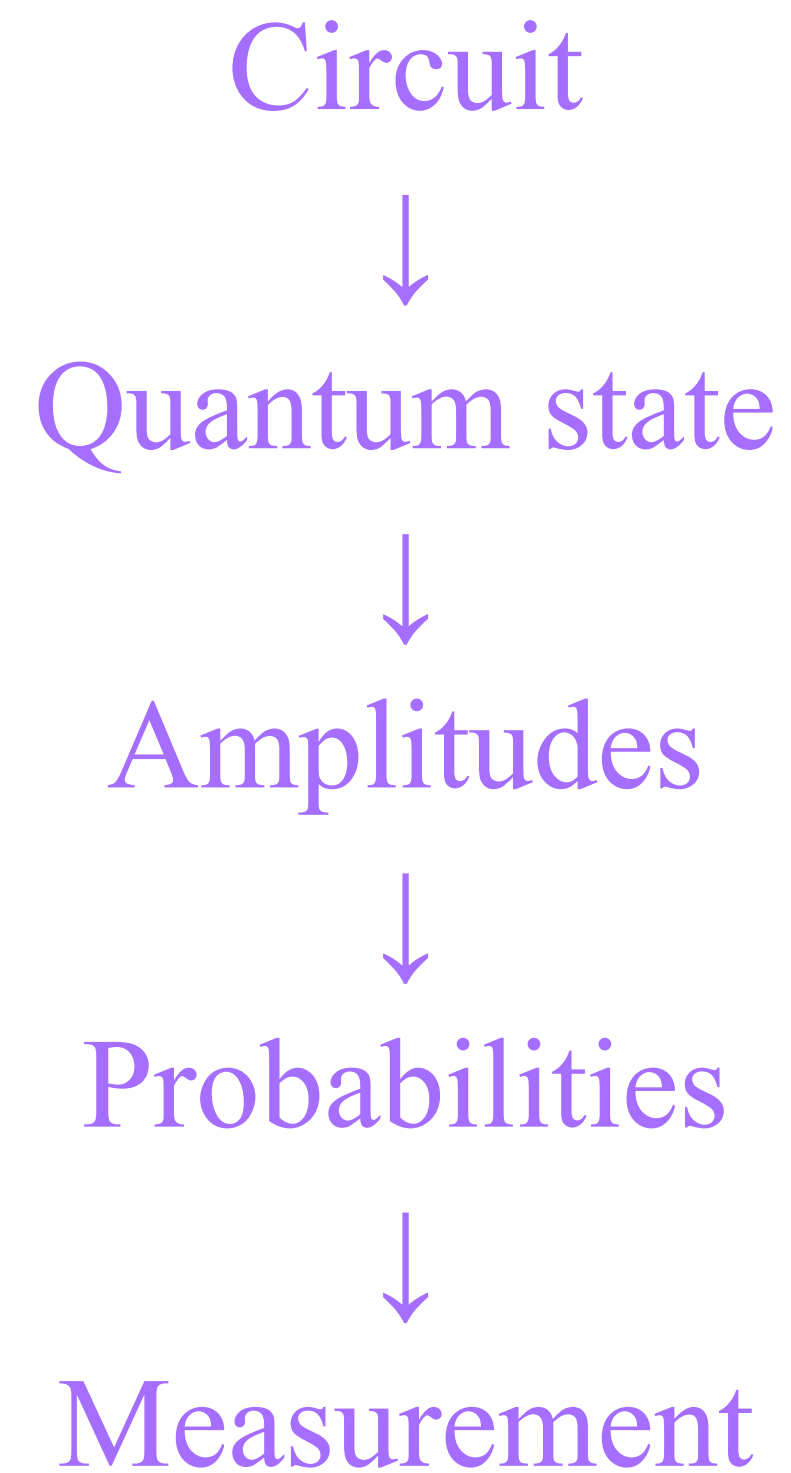
basis order: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$



For $a|00\rangle+b|01\rangle+c|10\rangle+d|11\rangle$: separable iff $ad = bc$.

The analysis workflow used in the notebook

- 1 Build circuit
- 2 `Statevector.from_instruction()`
- 3 `probabilities_dict()`
- 4 `measure_all()`
- 5 Run shots
- 6 Compare prediction ↔ observation



predict ↔ observe

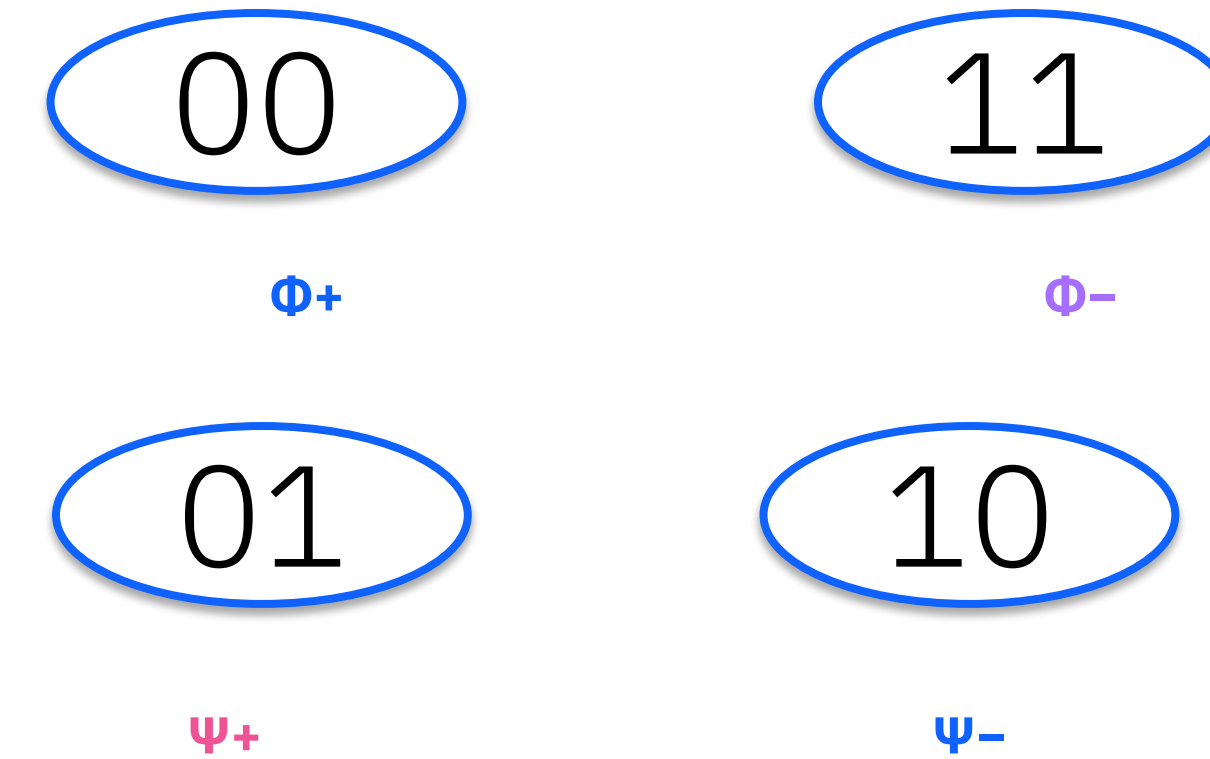
Four Bell states

$$\Phi^+ = (|00\rangle + |11\rangle)/\sqrt{2}$$

$$\Phi^- = (|00\rangle - |11\rangle)/\sqrt{2}$$

$$\Psi^+ = (|01\rangle + |10\rangle)/\sqrt{2}$$

$$\Psi^- = (|01\rangle - |10\rangle)/\sqrt{2}$$



four states • two bits of structure

Φ vs $\Psi \rightarrow$ basis components

$+$ vs $- \rightarrow$ relative phase

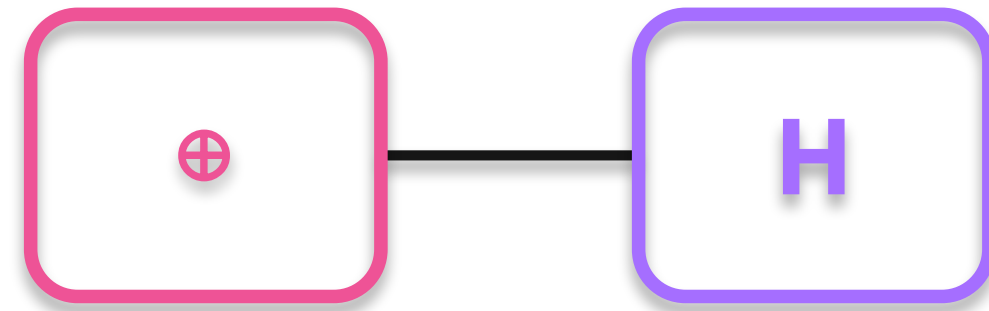
Why an analyzer is needed

DIRECT Z-BASIS MEASUREMENT

Φ^+ and $\Phi^- \rightarrow 00$ or 11

Ψ^+ and $\Psi^- \rightarrow 01$ or 10

Phase sign is invisible here.



decode phase $\rightarrow$ basis label

INVERSE BELL TRANSFORM

$CX(q_0, q_1)$

$\downarrow$

$H(q_0)$

$\downarrow$

Measure

Phase information becomes a basis
label.

From Bell states to GHZ

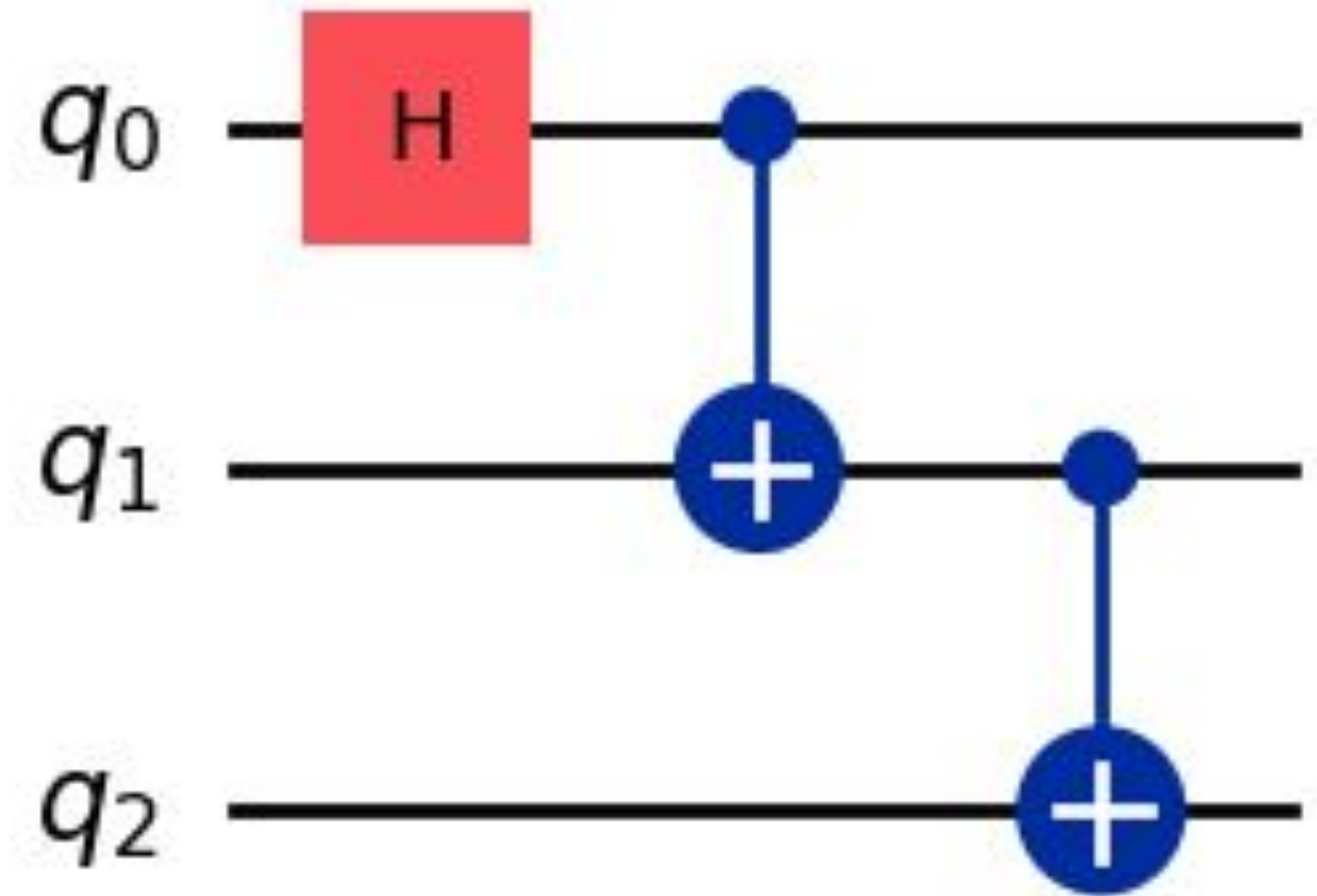
$|000\rangle$

↓ $H(q_0)$

$(|000\rangle + |001\rangle)/\sqrt{2}$

↓ $CX(q_0, q_1), CX(q_1, q_2)$

$|GHZ\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$



Three qubits → 8 amplitudes; only 000 and 111 are non-zero.

GHZ vs independent superposition

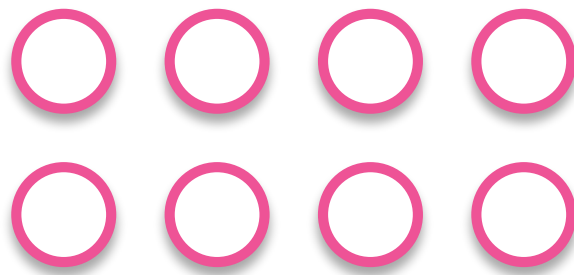
GHZ
 $(|000\rangle + |111\rangle)/\sqrt{2}$

2 non-zero amplitudes
 $000 \approx 50\%$
 $111 \approx 50\%$

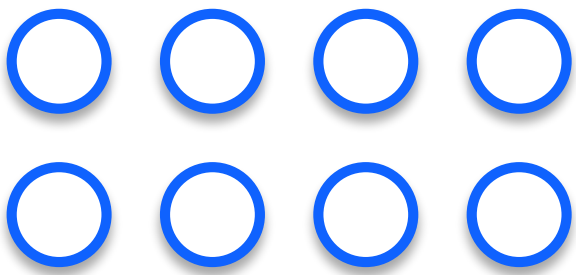
INDEPENDENT H
 $|+\rangle|+\rangle|+\rangle$

8 non-zero amplitudes
Each outcome $\approx 12.5\%$

Not entangled



GHZ



H⊗H⊗H

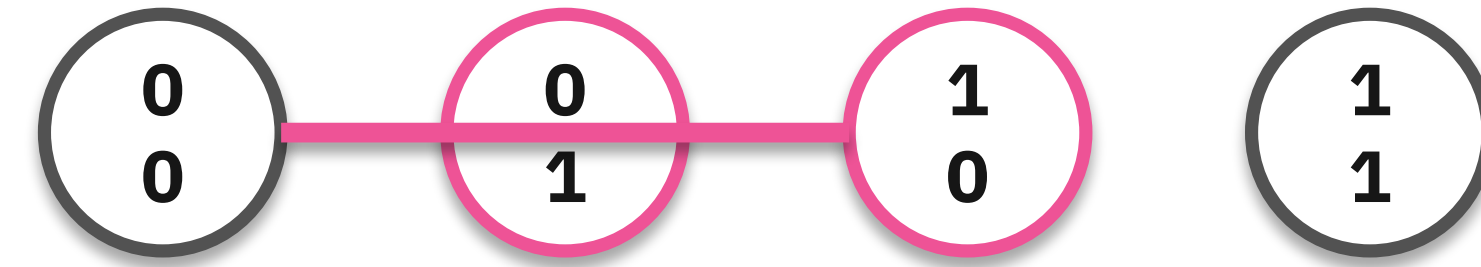
2 active amplitudes vs 8

Predict before
you build

TARGET

$$|\psi\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$$

Then generalize to
 $(|001\rangle + |110\rangle)/\sqrt{2}$



predict non-zero states first

PREDICT

Statevector

Non-zero amplitudes

Probabilities

Correlation

Entanglement

DESIGN

Choose gates

Construct circuit

Verify with

Statevector

Measure with shots

Explain the result

Your hands-on roadmap



build • inspect • measure • explain

01

Create

2- and 3-qubit circuits

02

Control

Use X, H and CNOT

03

Analyze

Statevectors +
probabilities

04

Distinguish

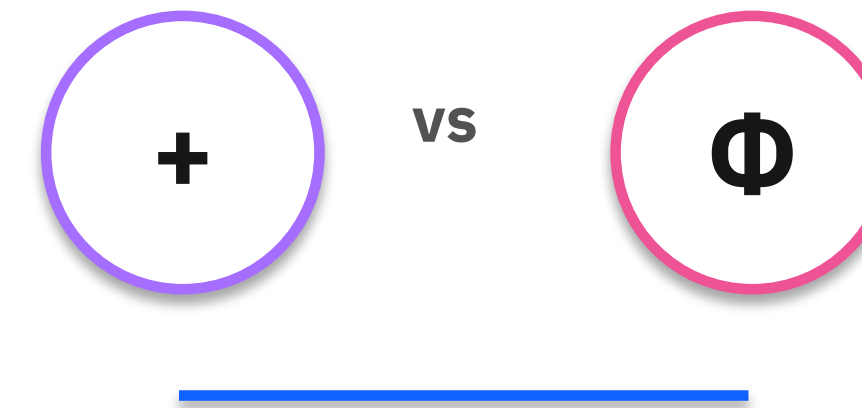
Product vs entangled
states

Ready for the lab?

Can you explain why

$$|+\rangle|+\rangle \neq |\Phi^+\rangle$$

and predict the measurement outcomes
before running the circuit?



predict before you execute

If yes, open Lab 02 and start with prediction—not execution.